

InsightAI Executive Report

Based on the analysis of 1000 records, the data presents a robust 99.6% health index. We recommend a classification model targeting 'Churn' using the available 6 numeric features. Immediate opportunity exists to clean remaining minor flaws and pilot a baseline model to drive measurable business KPIs.

Dataset Overview

Dataset contains 1,000 rows and 10 columns. 6 numeric fields, 3 categorical fields, and 0 datetime-like fields identified. Identifier columns excluded from analysis: Customer_ID. Health Score: 99.6% (Completeness: 99.6%, Duplicate Penalty: 0.0%).

Health Score

99.6

Key Findings

- The dataset features 1000 rows across 10 columns with 99.6% health score.
- Tenure_Months and Total_Charges show the strongest correlation at 0.77.
- Categorical analysis reveals 'Month-to-month' is the dominant segment in Contract_Type (520 occurrences, 52.0% of total).
- Model target candidates include Churn.

Business Opportunities

- With 6 numeric features available, the strong correlation (0.77) between Tenure_Months and Total_Charges suggests these variables could be combined or used as interaction terms in modeling.
- Design targeted interventions for the dominant 'Month-to-month' cohort (52.0% of total) to maximize impact within the Contract_Type segment.
- With 17.4% of records labeled '1', predictive modeling for 'Churn' must incorporate class balancing techniques.

Risk Factors

- Missing values detected in 3 columns, leading to possible training biases.
- Anomalous values detected: Monthly_Charges (5 outliers), Total_Charges (3 outliers) (IQR method) could distort predictions.
- Framing as classification requires valid stakeholder alignment on performance threshold.

ML Strategy

Primary path: classification workflow with Logistic Regression as leading candidate.